

Individual Reflective Piece – Network Security Module

What? (Summary of Learning Experience)

This Network Security module offered a hands-on exploration of vulnerability assessments, security standards, and responsible penetration testing. In Assessment 1, I audited demo.testfire.net using tools like Nmap, Nikto, and OpenVAS, identifying web-based vulnerabilities and assessing baseline risks. However, my tutor's feedback pointed out an overemphasis on OWASP Top 10 and insufficient attention to system-level or middleware issues.

This helped me improve in Assessment 2. I adopted a layered approach and used tools like Burp Suite, OWASP ZAP, SSLyze, WHOIS, and DIG to uncover issues such as deprecated TLS use, SMB port exposure, and DNS misconfigurations. I also mapped findings to GDPR and PCI DSS standards, improving my awareness of how vulnerabilities relate to legal and regulatory obligations (European Union, 2016; PCI SSC, 2022).

Through these assessments, I developed a more holistic method, learning to balance surface-level issues with architectural weaknesses and compliance relevance. I also created a mental checklist to guide future assessments, making my approach more structured and mindful of real-world scenarios.

So What? (Reflection and Analysis)

Initially, I approached the module with a checklist mindset, identifying vulnerabilities and reporting them. However, the feedback made me realise that a true cybersecurity professional must think beyond surface-level issues. It is not enough to report a missing security header; one must understand its impact on business, compliance, and user safety.

My perspective shifted when I reviewed the system architecture holistically and connected each tool's findings to security standards. For instance, identifying a deprecated TLS version through SSLyze was not just a technical observation, it also constituted a breach of GDPR Article 32, which requires robust data encryption (European Union, 2016). Understanding such legal ramifications deepened my appreciation for the ethical and professional responsibility involved in cybersecurity work.

I also learned to work more independently. Although support was available, much of the troubleshooting, especially configuring Burp Suite and proxy settings inside Kali Linux was done by self-learning and testing. This boosted my confidence and reaffirmed my technical problem-solving skills.

Peer discussions and tutor insights played an important role in enhancing my critical thinking. Observing other's approaches helped me question the depth of my own. I also improved my time management by creating a structured plan for each phase of the assessment, ensuring that scans, screenshots, and analysis were completed progressively without last-minute stress.

I also experienced a mindset shift, moving from just 'completing an assignment' to genuinely engaging with the security implications of my actions. This emotional investment made the learning feel more impactful and helped me connect academic tasks with real-world responsibilities in the cybersecurity domain.

Now What? (Application and Development)

Going forward, I understand that cybersecurity assessments should always balance technical insight with business relevance. Tools are valuable, but context, interpretation, and prioritisation determine whether the findings are actionable. I plan to continue sharpening my ability to analyse vulnerabilities not just as code flaws, but as potential legal, ethical, and reputational risks for organisations.

Through this module, I identified gaps in my skills and used a professional development plan (PDP) to address them. One area I want to focus on is encryption. I plan to explore how TLS, SSL certificates, and secure key management operate in

enterprise environments. I'm also interested in learning SIEM platform like Splunk, which play a critical role in real-time monitoring and log analysis (Chapple et al., 2022).

Professionally, I now feel more prepared to contribute as a security analyst or vulnerability assessor. This reflection has made me realise the importance of maintaining ethical boundaries, respecting legal limitations in scanning, and communicating clearly in a professional tone.

Looking ahead, I plan to participate in Capture the Flag (CTF) events and contribute to security forums to stay engaged with the community. I'll also keep myself updated through threat intelligence sources to stay informed of new exploits, vulnerabilities, and industry responses.

Personal Skills Developed

- Time Management: I paced my work across both assessments, avoiding cramming and leaving time for reviews and editing.
- Critical Thinking: I learned to go beyond basic scan outputs and interpret findings from a compliance and business lens.
- Digital Literacy: Mastered Kali Linux VM operations, proxy setups, and command-line scanning tools.
- Research: I frequently referred to tool documentation and GDPR/PCI DSS standards to justify findings.
- Problem-Solving: Encountered and resolved tool limitations and configuration issues independently.
- Ethical Awareness: Understood and respected boundaries in penetration testing, avoiding destructive techniques or scope violations.

Conclusion

This module has been a transformative journey in my cybersecurity education. I moved from a narrow, tool-based mindset to a more strategic, compliance-aware approach. I am now more confident in my technical and professional capabilities and feel equipped to contribute meaningfully to a security team. I also understand that continuous learning, ethical conduct, and effective communication are essential pillars of a successful career in cybersecurity.

References

Chapple, M., Stewart, J. and Gibson, D. (2022) CISSP (ISC)2 Certified Information Systems Security Professional Official Study Guide. 9th edn. Hoboken: Wiley.

European Union (2016) Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32016R0679> (Accessed: 4 June 2025).

PCI SSC (2022) PCI DSS Quick Reference Guide: Understanding the Payment Card Industry Data Security Standard version 4.0. Available at: https://www.pcisecuritystandards.org/documents/PCI-DSS-v4_0-Quick-Reference-Guide.pdf (Accessed: 4 June 2025).